1 基礎運算 Basic Operation

$$a+b=b+a$$
$$a+(b+c)=(a+b)+c$$
$$a\cdot b=b\cdot a$$
$$a\cdot(b\cdot c)=(a\cdot b)\cdot c$$
$$a\cdot(b+c)=a\cdot b+a\cdot c$$

$$a+(b\pm c)=a+b\pm c$$
$$a-(b\pm c)=a-b\mp c$$

2 分數和比 Fraction and Ratio

$$\frac{a}{c}+\frac{b}{c}=\frac{a+b}{c}\ (c\neq 0)$$
$$\frac{a}{c}+\frac{b}{d}=\frac{ad+bc}{cd}\ (c\neq 0,d\neq 0)$$
$$a\cdot\frac{b}{c}=\frac{ab}{c}\ (c\neq 0)$$
$$\frac{a}{c}\cdot\frac{b}{d}=\frac{ab}{cd}\ (c\neq 0,d\neq 0)$$
$$\frac{ab}{ac}=\frac{b}{c}\ (a\neq 0,c\neq 0)$$
$$a\frac{b}{c}=\frac{ac+b}{c}\ (c\neq 0)$$
$$\frac{a}{c}\div\frac{b}{d}=\frac{a}{c}\cdot\frac{d}{b}=\frac{ad}{bc}\ (b\neq 0,c\neq 0)$$

$$a:b=a\div b=\frac{a}{b}\ (b\neq 0)$$
$$ac:bc=a:b=\frac{a}{d}:\frac{b}{d}\ (b\neq 0,c\neq 0,d\neq 0)$$
$$a:b=c:d\Leftrightarrow ad=bc\ (b\neq 0,d\neq 0)$$
$$\frac{a}{b}=\frac{c}{d}=\frac{a+b}{c+d}=\frac{a-b}{c-d}=\frac{a+b}{a-b}=\frac{c+d}{c-d}$$
$$\frac{a}{100}=a\%$$

3 乘方運算 Exponentiation

$$a^m \cdot a^n = a^{m+n}\ (a \neq 0)$$

$$(a^m)^n = a^{mn}\ (a \neq 0)$$

$$a^m b^m = (a \cdot b)^m\ (a \neq 0)$$

$$\frac{a^m}{b^m} = (\frac{a}{b})^m$$

4 根式運算 Radical operation

$$a^2 = b \Leftrightarrow\ a = \pm\sqrt{b}\ (b \geq 0)$$

$$\sqrt{a^2} = a\ (a \geq 0)$$

$$\sqrt{a^2} = -a\ (a \leq 0)$$

$$(\sqrt{a})^2 = a (a \geq 0)$$

$$\sqrt{a^2 b} = \sqrt{a^2} \cdot \sqrt{b} = a\sqrt{b}\ (a \geq 0, b \geq 0)$$

5 絕對值運算 Absolute value operations

$$|-x| = |x|$$

$$|x - y| = |y - x|$$

$$|xy| = |x| \cdot |y|$$

$$\left|\frac{x}{y}\right| = \frac{|x|}{|y|}, y \neq 0$$

6 代數式運算 Algebraic operations

$$x^2 - y^2 = (x + y)(x - y)$$

$$(x \pm y)^2 = x^2 \pm 2xy + y^2$$

7 等差數列 Arithmetic sequence

$$d = a_{n+1} - a_n$$

$$a_n = a_1 + (n-1)d$$

$$S_n = \frac{(a_1 + a_n)n}{2}$$

$$\mathrm{d} = \frac{a_\mathrm{m} - a_\mathrm{n}}{\mathrm{m} - \mathrm{n}}$$

8 等比數列 Geometric Sequences

$$\mathrm{q} = \frac{a_{n+1}}{a_n}$$

$$a_n = a_1 \times q^{n-1} = a_m \times q^{n-m}$$

9 必背計算結論 Calculation Conclusion

$$2^0 = 1$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

$$2^6 = 64$$

$$2^7 = 128$$

$$2^8 = 256$$

$$2^9 = 512$$

$$2^{10} = 1024$$

$$2^{11} = 2048$$

$$2^{12} = 4096$$

$$2^{13} = 8192$$

$$2^{14} = 16384$$

$$2^{15} = 32768$$

$$2^{16} = 65536$$

$$11^2 = 121$$
$$12^2 = 144$$
$$13^2 = 169$$
$$14^2 = 196$$
$$15^2 = 225$$
$$16^2 = 256$$
$$17^2 = 289$$
$$18^2 = 324$$
$$19^2 = 361$$

$$15^2 = 225$$
$$25^2 = 625$$
$$35^2 = 1225$$
$$45^2 = 2025$$
$$55^2 = 3025$$
$$65^2 = 4225$$
$$75^2 = 5625$$
$$85^2 = 7225$$
$$95^2 = 9025$$

$$1! = 1$$
$$2! = 2$$
$$3! = 6$$
$$4! = 24$$
$$5! = 120$$
$$6! = 720$$

$$\pi \approx 3.1416$$
$$e \approx 2.7183$$
$$\sqrt{2} \approx 1.4142$$
$$\sqrt{3} \approx 1.7321$$
$$\sqrt{5} \approx 2.2361$$
$$\sqrt{6} \approx 2.4495$$

$$3^2 + 4^2 = 5^2$$
$$5^2 + 12^2 = 13^2$$
$$7^2 + 24^2 = 25^2$$
$$8^2 + 15^2 = 17^2$$

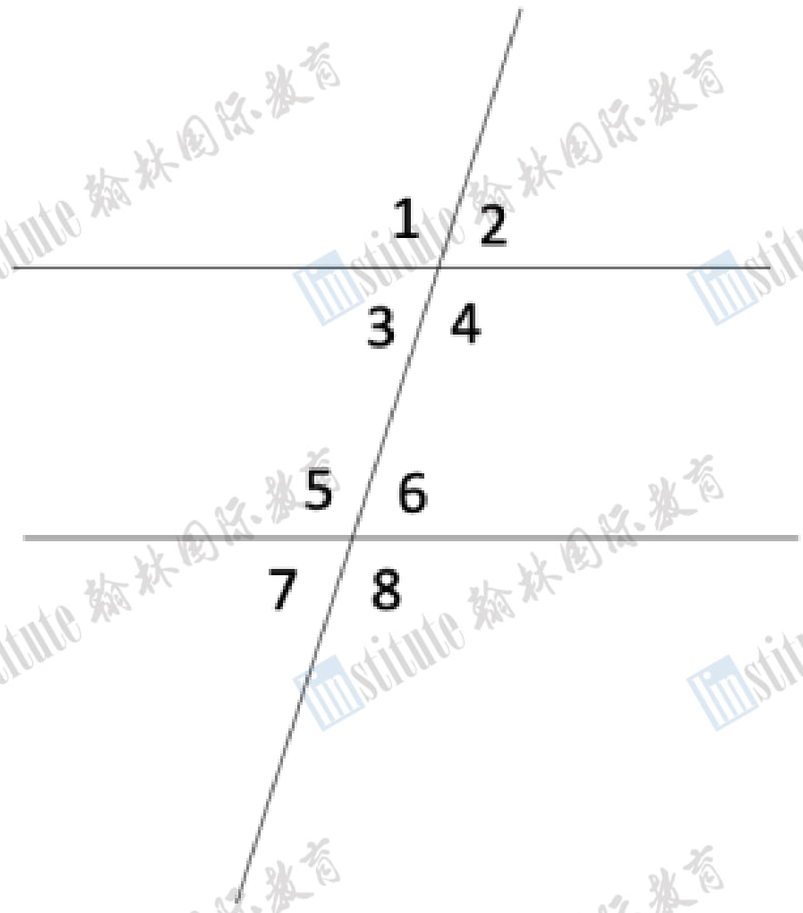
1 角和線 Angles and Lines

- Sum of complementary angles: a+b=90°

  互余的兩個角和為90°

- Sum of supplementary angles: a+b=180°

  互補的兩個角和為180°



- Relationship of Angles Formed by Parallel Lines:

  平行線間角的關系（“三線八角”）：

  Alternate interior angles: $\angle 3 = \angle 6, \angle 4 = \angle 5$

  内錯角相等

  Alternate exterior angles: $\angle 1 = \angle 8, \angle 2 = \angle 7$

  外錯角相等

  Corresponding angles: $\angle 1 = \angle 5, \angle 2 = \angle 6, \angle 3 = \angle 7, \angle 4 = \angle 8$

  同位角相等

  Interior angles on the same sides of transversal:

  $\angle 3 + \angle 5 = 180°, \angle 4 + \angle 6 = 180°$

  同旁内角互補

- 三角形的性質 The Property of Triangles

  The area of a triangle $= \frac{1}{2} \times$ base $\times$ height

  三角形的面積 $= \frac{1}{2}$ 底 $\times$ 高

  The sum of any two sides in a triangle is greater than the third side, and the difference be-tween any two sides is less than the third side. 三角形兩邊之和大於第三邊，兩邊之差小於第三邊。

- Pythagoras Theorem: $a^2 + b^2 = c^2$

  勾股定理：直角三角形中，兩直角邊的平方和等於斜邊的平方

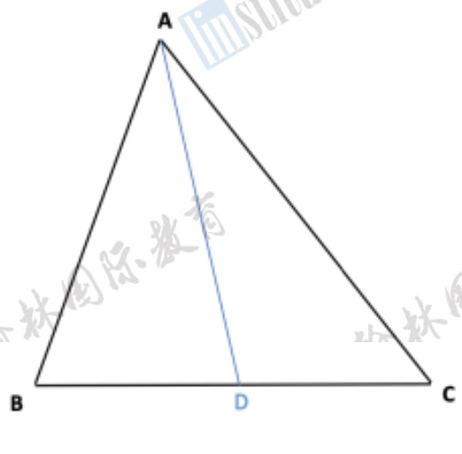
  (*常考勾股数：* $3^2 + 4^2 = 5^2, 5^2 + 12^2 = 13^2, 7^2 + 24^2 = 25^2, 8^2 + 15^2 = 17^2$)

- Mid-line of triangle:

  三角形中線性質

  BD = DC,

  $S_{\Delta ABD} = S_{\Delta ADC}$

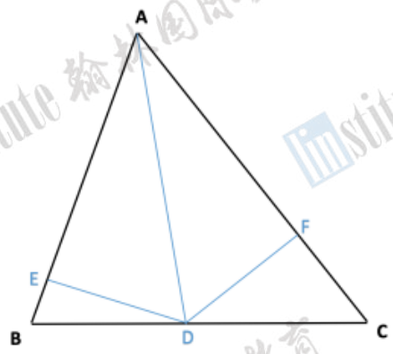


- Triangle bisector:

  三角形角平分線性質

  $\angle BAD = \angle CAD$

  DE = DF

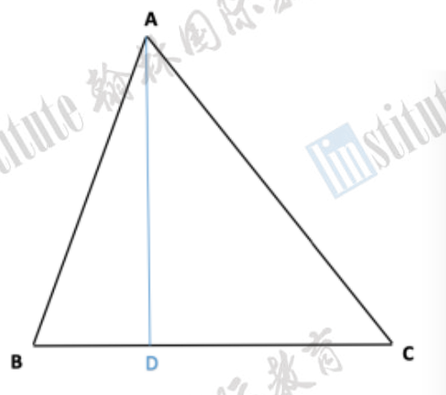


- Altitude of triangle:

  三角形垂線性質

  AD ⊥ BC

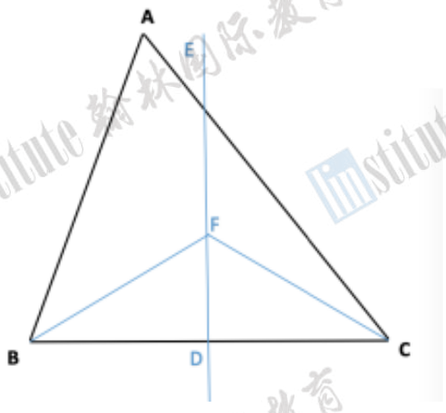
  $S_{\Delta ABC} = \frac{1}{2} \times AD \times BC$



- Vertical bisector of a triangle:

  三角形垂直平分線性質

  BD=DC, DE ⊥ BC, FB=FC

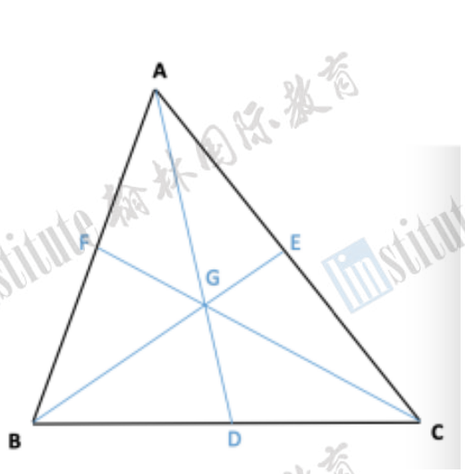


- Center of gravity of triangle:

  三角形重心性質

  BD=DC, AE=EC, AF=FB

  AG=2GD, BG=2GE, CG=2GF



- Middle line of the triangle:

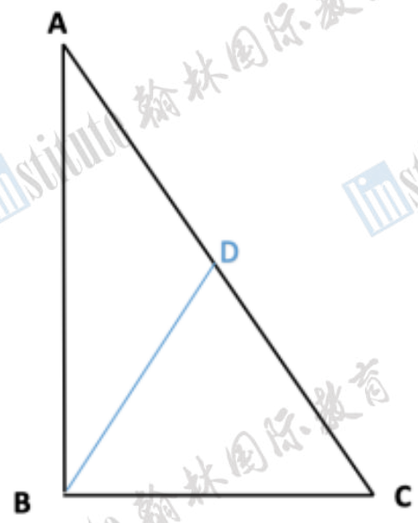
  三角形中位線性質

  $DE = \frac{1}{2} BC, DE // BC$

- **Isosceles triangle:** Same sides same angles; Three lines in one.
  等腰三角形兩腰相等，兩底角相等，三線（中線、角平分線、垂線）合一。

  Equilateral triangle: Same side length, same angles; $\text{Area} = \frac{\sqrt{3}}{4}a^2 (a = \text{side length})$
  等邊三角形各邊、各角都相等，面積= $\frac{\sqrt{3}}{4}$ 邊長的平方。

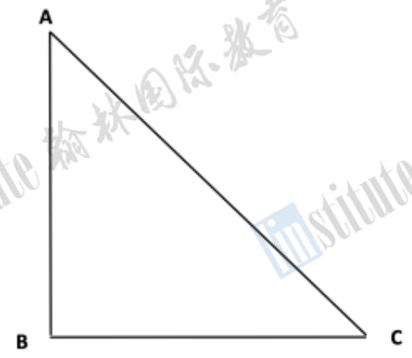
  Right triangle: BD= $\frac{1}{2}$ AC
  直角三角形斜邊中線等於斜邊的一半



- **Isosceles right triangle:**

$$AB = BC = \frac{\sqrt{2}}{2}AC$$

$$\angle BAC = \angle BCA = 45°$$



等腰直角三角形斜邊是兩直角邊的 $\sqrt{2}$ 倍，兩銳角都是45°.

2 相似三角形 Similar Triangles

- **Determination of Similar Triangles相似三角形的判定**

Principle 1 (SSS)

方法一：三邊對應成比例

If $\frac{AB}{A'B'} = \frac{BC}{B'C'} = \frac{CA}{C'A'}$, then $\Delta ABC \sim \Delta A'B'C'$.

If $\frac{AB}{A'B'} = \frac{BC}{B'C'}$, and $\angle B = \angle B'$, then

$\Delta ABC \sim \Delta A'B'C'$.

(AND VICE VERSA)

（反之，如果兩個三角形相似，以上邊的比例關系和角的相等關系也成立）

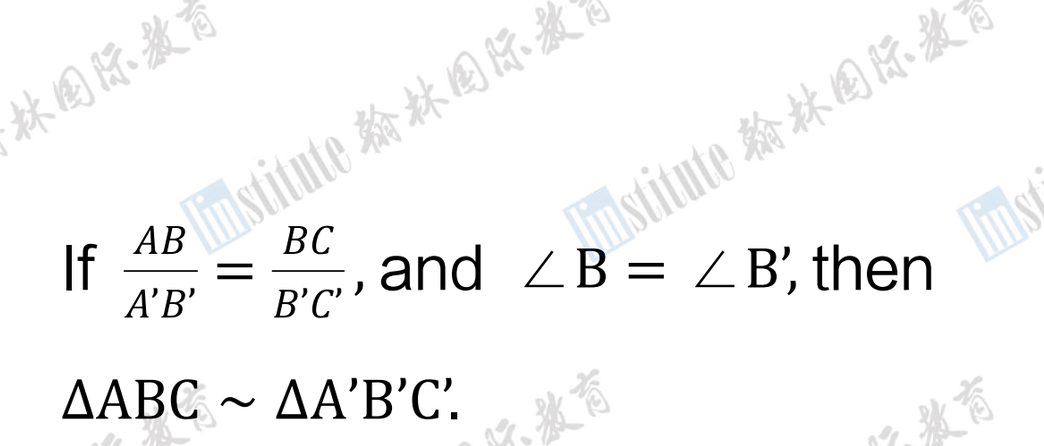
Principle 2 (AA)

方法二：兩角對應相等

If $\angle A = \angle A'$, $\angle B = \angle B'$, then $\Delta ABC \sim \Delta A'B'C'$.

Principle 3 (SAS)

方法三：兩邊對應成比例，它們的夾角對應相等



幾何 Geometry

- **The properties of similar triangles 相似三角形的性質**

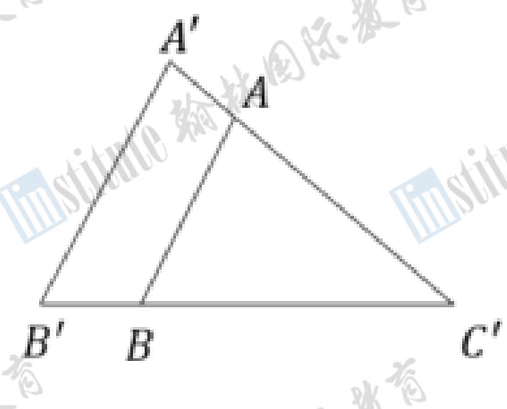
（1）The ratio of corresponding sides of similar triangles is called scale factor. The ratio of perimeters of similar triangles is equal to the scale factor. The ratio of areas of similar triangles is equal to the square of the scale factor.

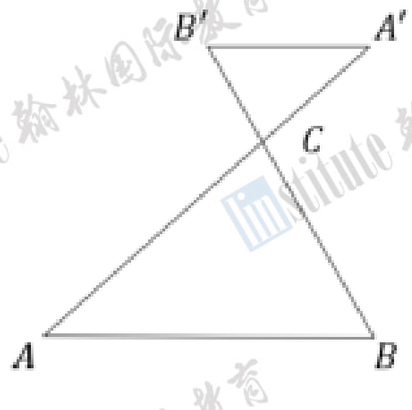
If $\Delta ABC \sim \Delta A'B'C'$, $\frac{AB}{A'B'} = \frac{BC}{B'C'} = \frac{CA}{C'A'} = k$, then $\frac{\text{perimeter of } \Delta ABC}{\text{perimeter of } \Delta A'B'C'} = k$, $\frac{\text{area of } \Delta ABC}{\text{area of } \Delta A'B'C'} = k^2$

相似三角形對應邊的比例稱為相似比。如果兩個三角形相似，那麼三邊對應成比例，並且**周長比等於相似比，面積比等於相似比的平方。**

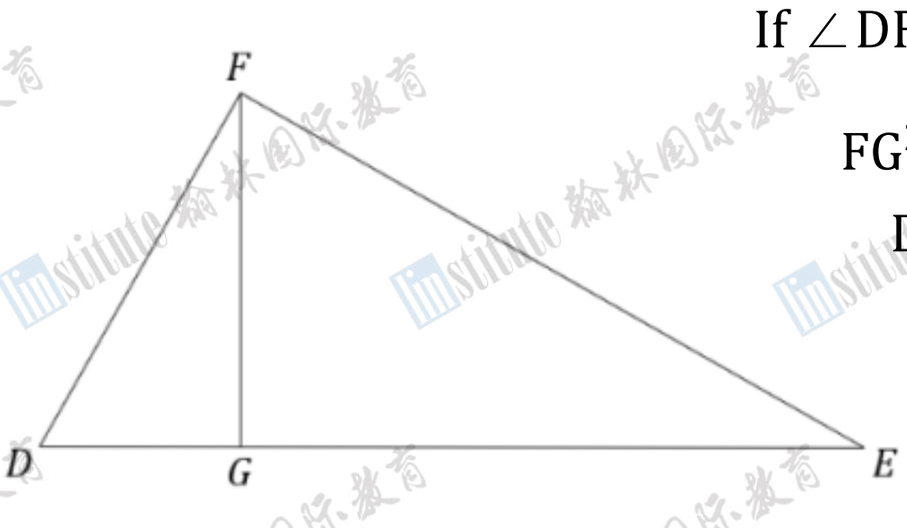
(2) A型和X型相似

If $AB \parallel A'B'$, then $\Delta ABC \sim \Delta A'B'C'$





(3) 相似三角形中的射影定理



If $\angle DFE = \angle FGE = 90°$, then $\Delta DFE \sim \Delta FGE$

$FG^2 = DG \times GE$, $FG \times DE = DF \times FE$

$DF^2 = DE \times DG$, $EF^2 = DE \times EG$

3 平行四邊形 Parallelograms

- **Properties of parallelograms:** opposite sides parallel and equal; opposite angles are equal; diagonals bisect each other.

平行四邊形的性質：對邊平行且相等，對角相等，對角線互相平分

- **Determinations of parallelograms:** a quadrilateral that has two pairs of opposite sides parallel / two pairs of opposite sides equal / one pair of opposite sides parallel and equal / two pairs of opposite angles equal / two diagonals bisect each other is parallelogram.

平行四邊形的判定：一個四邊形，如果它的兩組對邊分別平行/兩組對邊分別相等/一組對邊平行且相等/兩組對角分別相等/兩條對角線互相平分，那麼它是平行四邊形。

- **Properties of rectangles:** all angles = 90° ; diagonals are equal.

  矩形的性質：所有內角都是90°，兩條對角線相等

- **Properties of rhombus:** all sides are equal; diagonals bisect each other at 90° ; diagonals bisect angles

  菱形的性質：所有邊相等，對角線互相垂直平分，對角線平分內角

4 多邊形 Polygons

- A polygon with all its sides and all its angles equal is called a **regular polygon.**

  所有邊和角都相等的多邊形叫做**正多邊形**。

- **Number of symmetry axis** of a n-side regular polygon: n

  邊數為n的正多邊形有n條對稱軸。

- **order of rotational symmetry** of a n-side regular polygon: n

  邊數為n的正多邊形是n階旋轉對稱圖形

- **Sum of interior angles** of a n-side polygon = $(n-2)\times 180°$

  邊數為n的多邊形內角和為$(n-2)\times 180°$

- **Sum of exterior angles** of a n-side polygon=360°

  邊數為n的多邊形外角和等於360°

- **Number of diagonals** of a n-side polygon = $\frac{n(n-3)}{2}$

  邊數為n的多邊形的對角線個數為 $\frac{n(n-3)}{2}$

- **Interior angle** of a n-side regular polygon = $\frac{(n-2)\times 180°}{n}$

  邊數為n的正多邊形的內角大小為 $\frac{(n-2)\times 180°}{n}$

- **Exterior angle** of a n-side regular polygon = $\frac{360°}{n}$

  邊數為n的正多邊形的外角大小為 $\frac{360°}{n}$

- **Center angle** of a n-side regular polygon = $\frac{360°}{n}$

  邊數為n的正多邊形的中心角大小為 $\frac{360°}{n}$

5 圓 Circles

- Circumference of a circle = $\pi d = 2\pi r$

  圓周長= $\pi d = 2\pi r$

- Area of a circle = $\pi r^2 = \frac{1}{4}\pi d^2$

  圓面積= $\pi r^2 = \frac{1}{4}\pi d^2$

- **In the same circle or congruent circles,** if two central angles / circumference angles / arcs / chords / chord lengths on the two chords is equal, then the others are all equal.

  在同一個圓或全等的兩個圓中，如果兩個圓心角/兩個圓周角/兩條弧/兩條弦/兩段弦長相等，滿足其中一條，其它的也都成立。

- The degree of the circumference angle is equal to half of the degree of the central angle the arc it is facing.

  圓周角等於它所對的弧對應的圓心角的一半。

- For a sector, $\frac{arc\ length}{circumference} = \frac{central\ angle\ of\ the\ arc}{360°} = \frac{area\ of\ sector}{area\ of\ circle}$.

  扇形中，弧長在整圓周長中所佔的比，等於它所對的圓心角在360°中所佔的比，也等於它的面積在整圓面積中所佔的比。

6 解析幾何 Analytical Geometry

- The distance between two points: $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

  兩點間距離= $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

7 立體幾何 Solid Geometry

- For a prism, **V−E+F=2**: V stands for the number of vertices, E stands for the number of edges, F stands for the number of faces.

  在稜柱中，頂點數−稜數+面數=2.

8 立體幾何 Solid Geometry

- surface area of cuboid = $6a^2$

  正方體表面積：$6a^2$

- Volume of cuboid = a^3

  正方體體積：a^3

- Surface area of a prism = 2 x area of cross−section + perimeter of cross−section x length

  稜柱表面積 = 2倍橫截面面積+橫截面周長x稜柱的長

- Volume of a prism = area of cross−section x length

  稜柱體積=橫截面面積x稜柱的長

- Surface area of a cylinder = $2\pi rh + 2\pi r^2 = 2\pi r(h + r)$

  圓柱表面積 = $2\pi rh + 2\pi r^2 = 2\pi r(h + r)$

- Volume of a cylinder = $\pi r^2 h$

  圓柱體積 = $\pi r^2 h$

- Volume of a pyramid = $\frac{1}{3}$ × base area × perpendicular height

  正四稜錐體積 = $\frac{1}{3}$ 底面積 × 垂高

- Surface area of a cone = $\pi rl + \pi r^2 = \pi r(l + r)$

  圓錐表面積 = $\pi rl + \pi r^2 = \pi r(l + r)$

- Volume of a cone = $\frac{1}{3}\pi r^2 h$

  圓錐體積 = $\frac{1}{3}\pi r^2 h$

- Surface area of a sphere = $4\pi r^2$

  球表面積 = $4\pi r^2$

- Volume of a sphere = $\frac{4}{3}\pi r^3$

  球體積 = $\frac{4}{3}\pi r^3$

1 加法原理 Sum Rule(Additional Principle)

If an event E_1 can happen in n_1 ways, event E_2 can happen in n_2 ways, event E_k can happen in n_k ways, and if any event E_1, E_2, ... or E_k happens, the job is done, then the total ways to do the job is $n_1 + n_2 + \cdots + n_k$.

加法原理：如果一件事情有k種不同的做法，其中做法E_1有n_1種方式完成，做法E_2有n_2種方式完成，做法E_k有n_k種方式完成，並且做法E_1, E_2, ... or E_k都有可能出現，那麼這件事情的總完成方式共有$n_1 + n_2 + \cdots + n_k$種。

2 乘法原理 Product Rule(Multiplication Principle)

When a task consistes of k separate parts, if the first part can be done in n_1 ways, the second part can be done in n_2 ways, and so on through the kth part, which can be done in n_k ways, then total number of possible results for completing the task is given by the product: $n_1 \times n_2 \times n_3 \times \cdots \times n_k$.

如果一件事情分成k個步驟完成，其中第一步有n_1種方式完成，第二步有n_2種方式完成，直到第k步，有n_k種方式完成，那麼完成整件事情的方法數是 $n_1 \times n_2 \times n_3 \times \cdots \times n_k$.

3 概率的主要性質 Properties of probability

(1)The probability of an event is between 0 and 1. 任意事件的概率都在0到1之間。

(2)The probability of an impossible event is 0. 不可能事件的概率為0.

(3)The probability of a certain event is 1. 必然事件的概率為1.

The probability that an event will occur is equal to one minus the probability that it will not occur. 一件事情發生的概率等於1減去它不發生的概率。

排列組合 Permutation and Combination

$$n! = n \cdot (n-1) \cdots 3 \cdot 2 \cdot 1$$

正整數n的階乘n · (n−1) · · ·3·2·1

$$_nP_r = \frac{n!}{(n-r)!} \ (1 \le r \le n)$$

n個物體中選出r個排序，共有 $\mathbf{{}_nP_r = \frac{n!}{(n-r)!}}$ 種方法

$$ {}_nC_r = \frac{n!}{r!\,(n-r)!}\ (0 \le r \le n) $$

n個物體中選出r個，共有 $\mathbf{{}_nC_r = \frac{n!}{r!(n-r)!}}$ 種方法

$$ \frac{{}_nP_n}{p!\ \times\ q!\ \times\ r!\ \times \cdots} = \frac{n!}{p!\ \times\ q!\ \times\ r!\ \times \cdots}\ (p+q+r+\ldots = n) $$

n個物體排列，其中p個物體是一類，q個物體是另一類，r個物體是第三類，…，

共有 $\frac{{}_nP_n}{p! \times q! \times r! \times \cdots} = \frac{n!}{p! \times q! \times r! \times \cdots}$ 種排列方式

$$ \boldsymbol{{}_nC_0 + {}_nC_1 + \ldots + {}_nC_n = 2^n} $$

$$ Probability = \frac{number\ of\ ways\ that\ a\ certain\ outcome\ can\ occur}{total\ number\ of\ possible\ outcomes} $$

4 概率的基本公式

集合和韋恩圖 Sets and Venn Diagrams

$$ n(A \cup B) = n(A) + n(B) - n(A \cap B) $$

兩個事件的並集公式：集合A和集合B的並集中的元素數 = 集合A中的元素數 + 集合B中的元素數 − 集合A和集合B交集中的元素數

$$ n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C) $$

三個事件的並集公式：集合A、集合B和集合C的並集中的元素數 = 集合A中的元素數 + 集合B中的元素數 + 集合C中的元素數 − 集合A、集合B和集合C交集中的元素數

1 Odd and Even奇偶性
Odd是奇數，Even是偶數

even ± even = even

odd ± odd = even

even ± odd = odd

even x even = even

odd x odd = odd

even x odd = even

even ÷ even = even/odd/fraction

even ÷ odd = even/fraction

odd ÷ odd = odd/fraction

even ≠ odd

- The sum of two consecutive integers is odd: n+(n+1)=2n+1

  兩個連續整數的和是奇數

- The product of two consecutive integers is even: n × (n+1)=n(n+1)

  兩個連續整數的乘積是偶數

- Any two consecutive integers have opposite parity

  兩個連續整數有相反的奇偶性（一奇一偶）

- If the product of n positive integers is even, at least one of these n positive integers is even

  如果n個正整數的乘積是偶數，那麼這n個正整數中至少有一個是偶數

- If the product of n positive integers is odd, all of these n positive integers are odd

  如果n個正整數的乘積是奇數，那麼這n個正整數都是奇數

- If the number of odd integers is even, then the sum of them is even

  偶數個奇數的和是偶數

- If the number of odd integers is odd, then the sum of them is odd

  奇數個奇數的和是奇數

2 2的若干次方的整除性 Divisibility rule for 2, 4, 8, 16, and 2^n

A number is divisible by 2 if the last digit of the number is divisible by 2;

A number is divisible by 4 if the last 2 digits of the number is divisible by 4;

A number is divisible by 8 if the last 3 digits of the number is divisible by 8;

A number is divisible by 16 if the last 4 digits of the number is divisible by 16;

A number is divisible by 2^n if the last n digits of the number is divisible by 2^n.

判斷一個數能否被2, 4, 8, 16, … 2^n, 整除只要看這個數的最後1, 2, 3, 4, …, n位組成的數能否被2, 4, 8, 16, …, 2^n 整除

3 3和9的整除性 Divisibility rule for 3 and 9

A number is divisible by 3 if the sum of the digits of the number is divisible by 3;

A number is divisible by 9 if the sum of the digits of the number is divisible by 9.

判斷一個數能否被3或者9整除只要看這個數的數位之和能否被3或者9整除

4 5和5的若干次方的整除性 Divisibility rule for 5, 25, 125, and 5^n

A number is divisible by 5 if the last digit of the number is divisible by 5;

A number is divisible by 25 if the last 2 digits of the number is divisible by 25;

A number is divisible by 125 if the last 3 digits of the number is divisible by 125;

A number is divisible by 5^n if the last n digits of the number is divisible by 5^n.

判斷一個數能否被5, 25, 125, …, 整除只要看這個數的最後1, 2, 3, 4, …, n位組成的數能否被5, 25, 125, …, 整除

5 11的整除性 Divisibility rule for 11

If the difference between the sum of all digits in odd digit and the sum of all digits in even digit can be divised by 11, then the number is divisible by 11.

判斷一個數能否被11整除只要看這個數的所有奇數數位和偶數數位之和的差值能否被11整除

6 有多於等於2種不同質因數的合數的整除性
Divisibility rule for 6, 12, 14, 15, 18, 24

A number is divisible by 6 if the number is divisible by both 2 and 3;

A number is divisible by 12 if the number is divisible by both 3 and 4;

A number is divisible by 14 if the number is divisible by both 2 and 7;

A number is divisible by 15 if the number is divisible by both 3 and 5;

A number is divisible by 18 if the number is divisible by both 2 and 9;

A number is divisible by 24 if the number is divisible by both 3 and 8;

判斷一個數能否被一個至少有兩種質因數的合數整除只要把這個合數分解質因數後拆分成每個質因數的冪再分別判斷，如果都可以整除，那麼這個數就能被那個合數整除

7 分解質因數
Prime Factorization

- If a number can be written as $\boldsymbol{a = p_1^{b_1} p_2^{b_2} \ldots p_n^{b_n}}$, then the number of positive factor of is $(b_1+1)(b_2+1)\ldots(b_n+1)$: the sum of its factor is $(p_1^0 + p_1^1 + \ldots + p_1^{b_1})(p_2^0 + p_2^1 + \ldots + p_2^{b_2}) \ldots (p_n^0 + p_n^1 + \ldots + p_n^{b_n})$.

  一個整數的正因數數量等於 $(b_1+1)(b_2+1)\ldots(b_n+1)$:這些正因數的和為 $(p_1^0 + p_1^1 + \ldots + p_1^{b_1})(p_2^0 + p_2^1 + \ldots + p_2^{b_2}) \ldots (p_n^0 + p_n^1 + \ldots + p_n^{b_n})$.

- Define $[x]$is the largest integer less than or equal to . Then the power of prime number p does n ! have is $\left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \ldots$

  定義 $[x]$ 是小於等於x的最大整數，則里含有質因數P的數量為 $\left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \ldots$

8 最大公因數和最小公倍數
GCF and LCM

If (a,b) = d, and n is a positive integer, then (na, nb) = nd

(a,1) = 1; (a, a) = a; (a,b) = (b,a); (a,b) = (b, a−b)

$$GCF(a,b) \times LCM(a,b) = a \times b$$